



ANGELES UNIVERSITY FOUNDATION

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Logetik – Barangay Service Integration Platform

A Final Project
Presented to
The Faculty of the College of Computer Studies
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In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Information Technology

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Project Overview

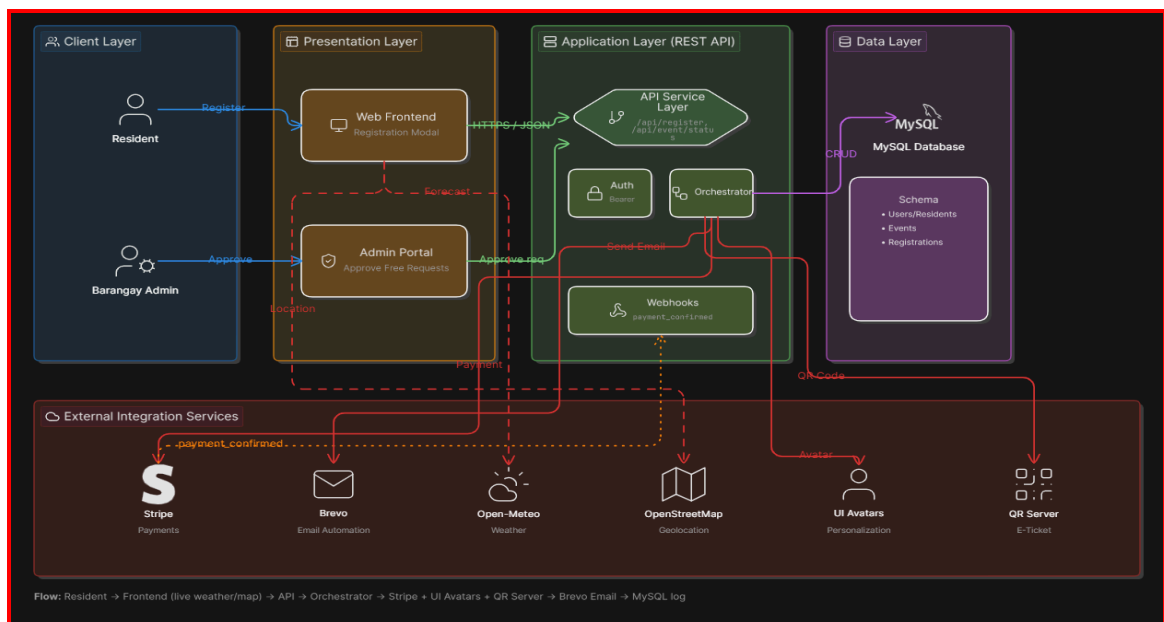
For this final project, Group Logetik has designed and implemented an integration solution for community services within a Barangay. While many barangay services currently operate with limited digital integration, Logetik acts as a centralized platform that bridges these gaps.

The primary objective is to implement an API-driven platform that improves the management of community programs—specifically focusing on automated registration, real-time data consumption, and secure financial transactions. The system targets the "Mushroom Farming" training program as its primary use case to demonstrate robust API-based communication.

System Architecture Documentation

The project follows a Layered (N-Tier) Architecture style, which allows for clean separation between user interface, business logic, and data persistence.

2.1 Architecture Diagram



The diagram should illustrate the flow from the Resident (Client) to the Web Server, the MySQL Database, and the various third-party API endpoints.

2.2 Component Description

- **Web Frontend:** Built to handle resident interactions, featuring an integrated registration modal that consumes live data for Guagua.

- **API Service Layer (Backend):** A REST API server that provides core services for the registration function.
- **Database System (MySQL):** Used for persistent storage of application data, resident records, and transaction logs
- **External Integration Services:** A suite of six specialized APIs (Stripe, Brevo, Open-Meteo, etc.) that handle automated workflows.

2.3 Data Flow Analysis

1. **Input:** Resident enters data into the registration form.
2. **External Fetch:** The system calls Open-Meteo and OpenStreetMap to provide environmental context and location services in real-time.
3. **Processing:** Upon submission, the Backend orchestrates calls to Stripe (for payments), UI Avatars (for personalization), and QR Server (for ticket generation).
4. **Output:** Brevo API compiles these elements into an automated, customized HTML email sent directly to the resident.

Architecture Analysis & Evaluation

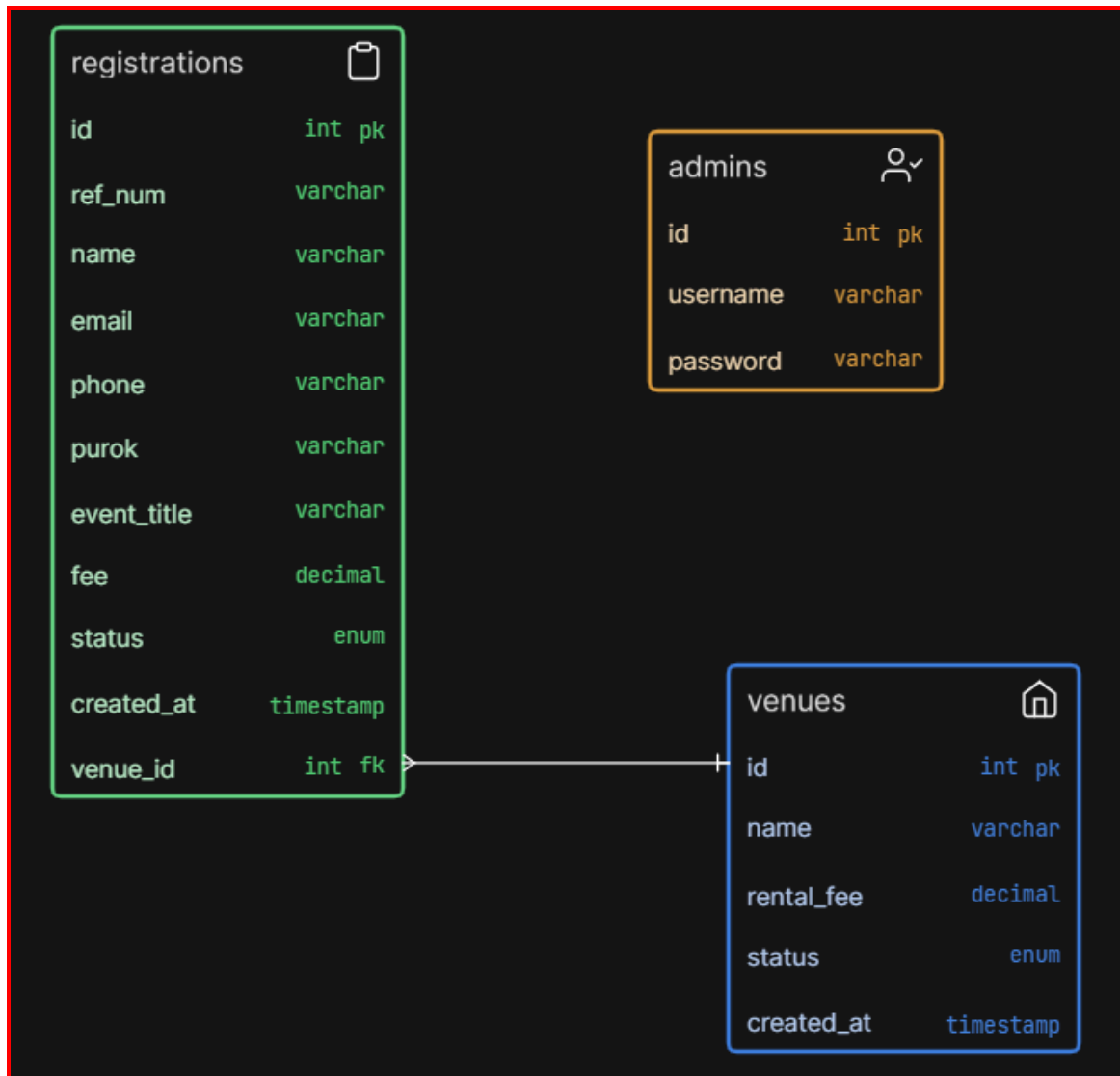
As part of the Course Learning Outcomes (CLO), we have analyzed the chosen architecture based on the following criteria:

- **Scalability:** By utilizing an API-centric design, the system offloads resource-heavy tasks (like QR code generation and email delivery) to external cloud providers. This reduces the load on the local Barangay server.
- **Security:** To protect resident data, the system uses Stripe for payment processing. This ensures that sensitive credit card information is handled by a PCI-compliant third party and never stored on our local MySQL database.
- **Limitations:** The solution is heavily reliant on internet connectivity. A local network failure would interrupt the connection to the Brevo and Stripe APIs, effectively halting the registration and notification workflow.

Database Schema

The following tables represent the core data entities required for the integration solution.

Table Name	Description	Key Fields
Users/Residents	Stores basic profile info for community members.	id, name, email, initials
Events	Stores details for programs like "Mushroom Farming."	event_id, title, fee, location_coord
Registrations	Links residents to events and tracks payment status.	reg_id, ref_number, status, qr_url



API Documentation

This section provides technical details for external systems to interact with the Loetik platform.

- **Base URL:** <https://api.loetik.auf.edu.ph/v1>
- **Authentication:** Bearer Token (Simulated for project scope).
- **Request/Response Format:** application/json.

Core Endpoints

1. **POST** /api/register

- **Purpose:** Registers a resident for an event.
- **Trigger:** Initiates the Brevo email and QR generation workflow.
- **Success Response:** 201 Created with the generated Reference Number.

2. **GET** /api/event/status

- **Purpose:** Checks the current registration count and weather forecast for a specific event.

3. **Error Codes:** 400 Bad Request (Validation errors), 401 Unauthorized (Invalid API key), 503 Service Unavailable (External API failure).

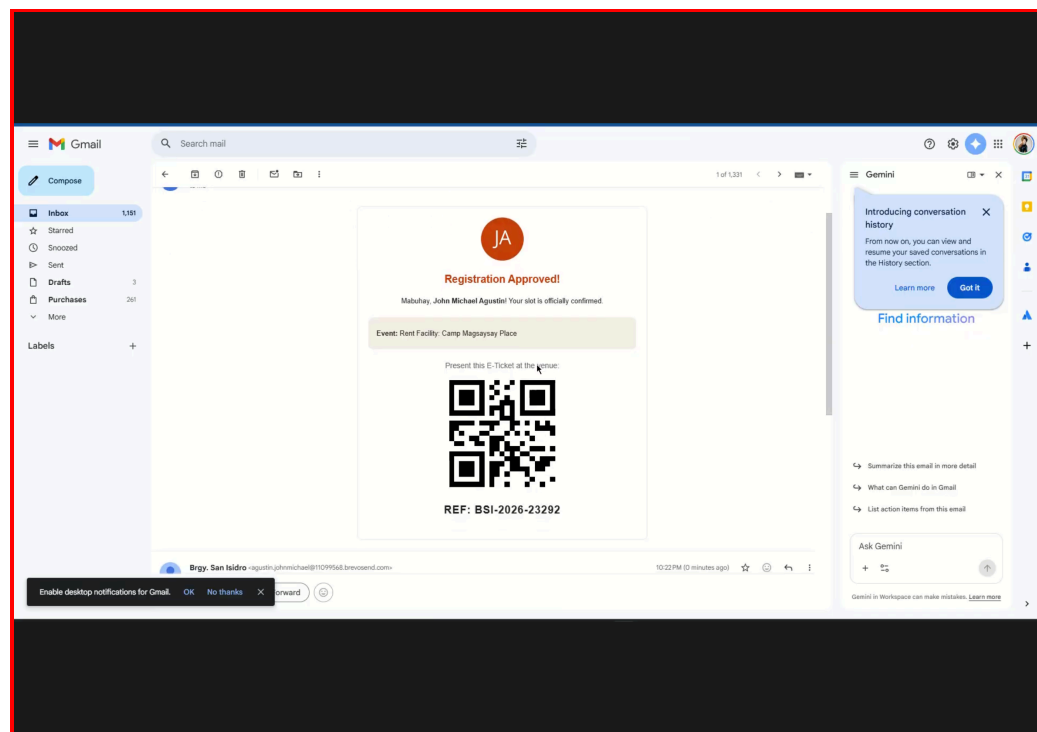
Integration Guidelines

To ensure seamless integration with external or future barangay modules, please follow these guidelines:

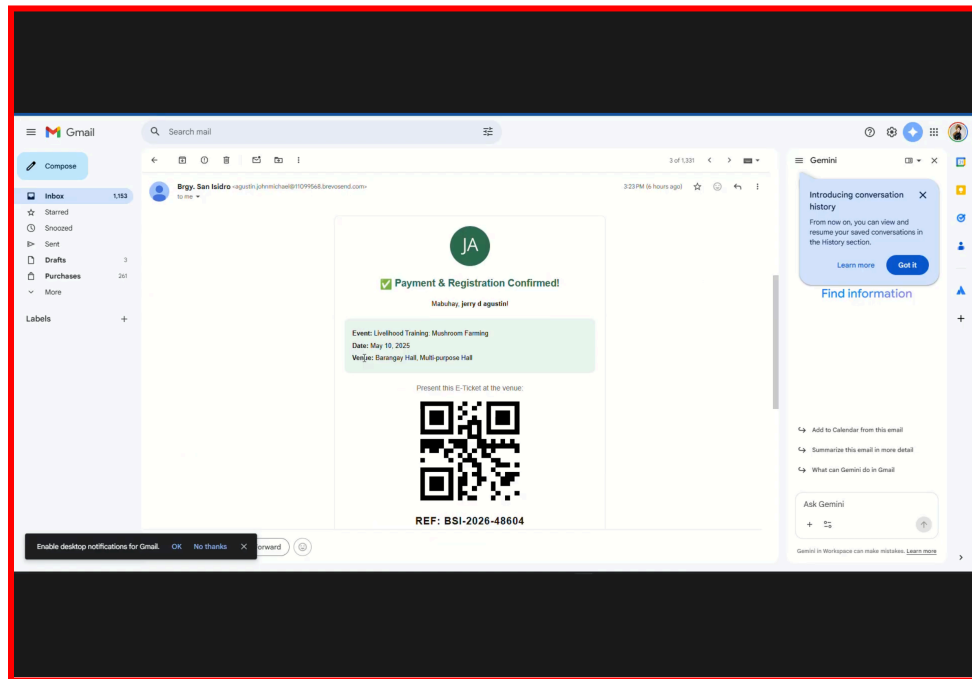
- **Webhooks:** External apps can listen for our payment_confirmed webhook to trigger secondary local processes.
- **Rate Limiting:** To prevent API exhaustion, external consumers are limited to 100 requests per hour.
- **Usage Example:** Use `curl -X POST -H "Content-Type: application/json" -d '{"event_id": "1", "name": "John Doe"}'` to test registration.

Testing and Evidence

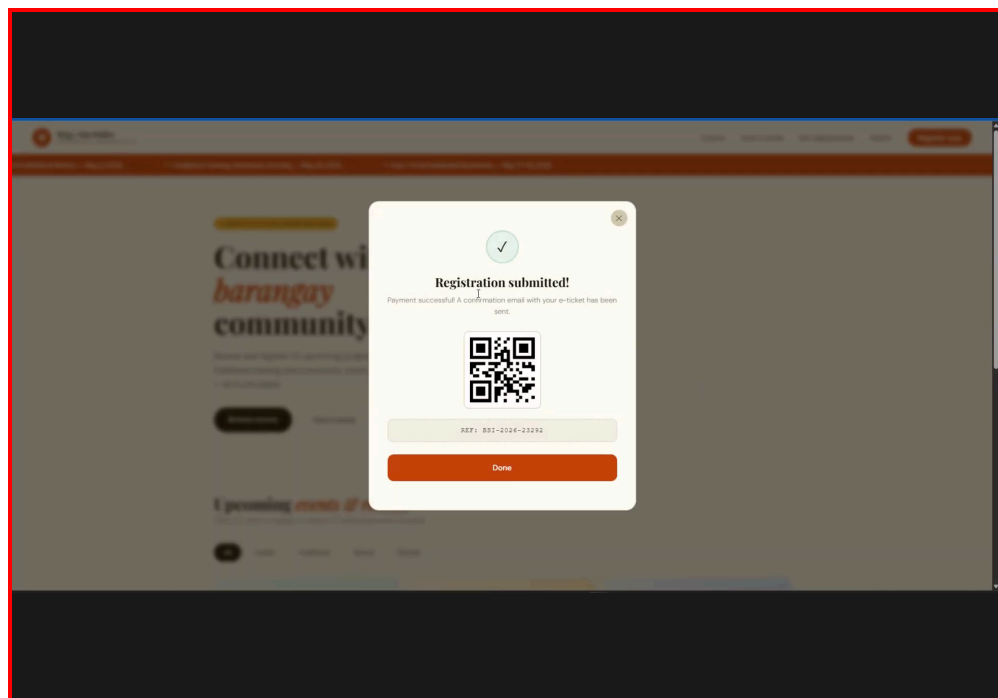
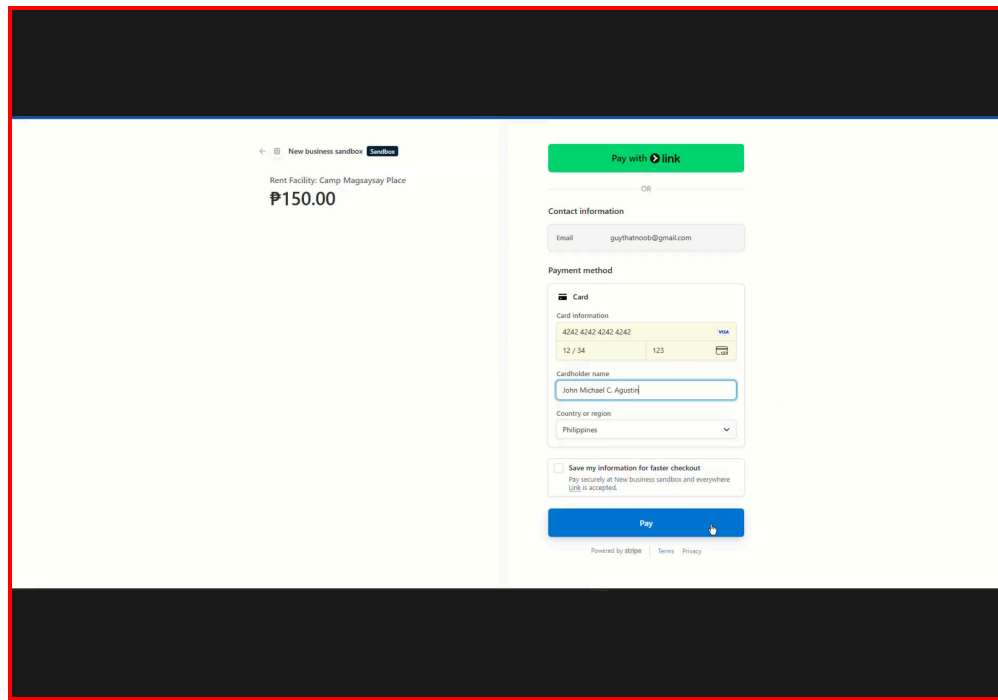
The following evidence demonstrates that the API services and automated workflows are functioning as designed.



Evidence: Postman screenshots showing successful API responses for registration.



Evidence: Screenshot of the confirmation email received via Brevo, showing the personalized UI Avatar and QR E-Ticket.



Evidence: Stripe Sandbox checkout screen and transaction confirmation.

Alignment with Learning Outcomes

This project fulfills the requirements of MCO2 by demonstrating the ability to:

1. **Analyze Integration Architectures:** Justifying the use of RESTful patterns and layered architecture.
2. **Design Solutions:** Creating the API contracts and UML diagrams for the Logetik platform.
3. **Implement Mechanisms:** Connecting six distinct external services into one cohesive barangay tool.
4. **Communicate Effectively:** Documenting the technical trade-offs and performance considerations in this report.